

LAB 2 [Group # → 9 to 16]

[Platform → HACKERRANK]

Introduction To Programming

(17-01-22)

QUESTIONS

Q1 → FIND DIVISORS

For a given number 'n', find all its divisors. A divisor of a number **n** is a positive integer that divides it without leaving a remainder.

Input Format

The first line contains a positive integer **n**.

Constraints

$$1 \leq n \leq 1000$$

Output Format

In subsequent lines, print divisors of n in increasing order.

Sample Input 0

```
12
```

Sample Output 0

```
1
2
3
4
6
12
```

Explanation 0

The factors of 12 are 1, 2, 3, 4, 6, 12. These factors have been printed out in each of the lines respectively.

Q2 → CAN SAM PLAY?

Sam likes playing computer games. But, playing computer games for a long time may affect Sam's health. So her mother restricted the number of minutes he could play.

On a general day, Sam can play up to **120** minutes (inclusive). Unless, if the day is a *weekend* (Saturday or Sunday), Sam is allowed to play up to **240** minutes (inclusive).

The days are encoded as follows :

- 0 -> Sunday,
- 1 -> Monday
- 2 -> Tuesday
- ...
- 5 -> Friday
- 6 -> Saturday

Sam will tell you the day encoded as above(**d**) and the number of minutes he wishes to play on that day(**t**), print “**YES**” if we can play following the restrictions set by his mom, otherwise print “**NO**”.

Input Format

The first line contains an integer '**d**', the encoded number for the day. The second line contains an integer '**t**', the number of minutes Sam wishes to play on the day **d**.

Constraints

$$0 \leq d \leq 6$$

$$1 \leq t \leq 300$$

Output Format

“**YES**” if Sam can play, otherwise “**NO**”

Sample Input 0

```
0
230
```

Sample Output 0

```
YES
```

Explanation 0

Since it is a weekend, Sam is allowed to play upto 240 minutes. Sam wished to play 230 minutes, he is allowed to do so. Hence, **YES** has been printed out.

Sample Input 1

```
4
200
```

Sample Output 1

NO

Explanation 1

It is not a weekend and sam is allowed to play for atmost 120 minutes. But he wishes to play game for 200 minutes. His mom would not allow so. Hence, *NO*

Q3 FIND MY BILL

I want to calculate my electricity bill. I know the number of **units** of electricity that has been used in my house. The government charges per unit are distributed as follows:

- For first 150 units, no charge
- For the next 100 units, Rs. 10 per unit is charged
- After 250 units, Rs. 15 per unit is charged

Given **N** number of units used by me, How much would I be charged?

Input Format

The first line contains an integer input, **N**

Constraints

$$0 \leq N \leq 1000$$

Output Format

The output is an *integer* denoting total charge in Rs for my electricity consumption.

Sample Input 0

180

Sample Output 0

300

Explanation 0

First 150 units are free. Remaining 30 units will be charged at Rs. 10 per unit. Therefore total bill amount is Rs. 300.

Sample Input 1

300

Sample Output 1

1750

Explanation 1

For the first 150 units, I will not be charged. Then for the next 100 units, my total charge would be $100 \times 10 = \text{Rs. } 1000$. The remaining 50 units would be charged at Rs. 15 per unit, for Rs. 750. Therefore my total bill will be $\text{Rs. } 1000 + \text{Rs. } 750 = \text{Rs. } 1750$.

Q4 WHAT'S YOUR BONUS

A company decided to give bonus to its employees according to the following criteria:

Time period of Service	Bonus
More than 8 years	15%
≥ 4 and ≤ 8 years	10%
Less than 4 years	5%

The bonus percentage is calculated on the salary of the employee.

The problem is to find the *bonus* amount for an employee given his salary and years of service in the company.

Input Format

First line contains an integer, *years of service*

Second line contains an integer, *salary*

Constraints

$20000 \leq \text{salary} \leq 500000$

$0 \leq \text{years of service} \leq 60$

Output Format

An integer.

Contains the bonus amount rounded to the greatest integer value smaller than the actual amount.(eg. if the actual bonus is 1222.57, then output should be 1222)

Sample Input 0

9

40000

Sample Output 0

6000

Explanation 0

Since *years of service* is greater than 8, the bonus would be 15 % of the salary. 15% of 40000 is 6000, which is the answer.

Q5 FIND X

Your friend had a problem he couldn't solve, so he asked you for help. Given **n** integers, return the value of **X**.

X is defined as the difference of count of even and odd numbers. (X = count of even numbers - count of odd numbers)

Input Format

The first line of input contains integer 'n'.

Each of next n lines contains a single integer .

Constraints

$1 \leq n \leq 100$

Output Format

Print a single integer **X**

Sample Input 0

```
5
1
2
3
4
5
```

Sample Output 0

```
-1
```

Explanation 0

2,4 are only even numbers. So count of even numbers = 2. 1,3,5 are odd numbers. So count of odd numbers = 3. $X = 2 - 3 = -1$

Q6 FIBONACCI SEQUENCE [NOT-GRADED]

Given a number **N**, print the N'th term of Fibonacci sequence.

Fibonacci sequence is a sequence in which each term is the sum of the previous 2 terms
i.e. 0,1,1,2,3,5,8,13

The first and second term in Fibonacci sequence is 0 and 1 respectively.

$$F[1] = 0, F[2] = 1$$

$$F[3] = F[1] + F[2] = 0 + 1 = 1 \Rightarrow \text{3rd term}$$

$$F[4] = F[2] + F[3] = 1 + 1 = 2 \Rightarrow \text{4th term}$$

Input Format

The first and only line contains a single integer 'N'

Constraints

$$1 \leq N \leq 100$$

Output Format

Output a single integer denoting N'th term of Fibonacci sequence

Sample Input 0

1

Sample Output 0

0

Explanation 0

1st term is 0

Sample Input 1

2

Sample Output 1

1

Explanation 1

2nd term is 1

Q7 LARGEST TWO [NOT GRADED]

Sam got a question in his IP lab, he solved it. Can you?

Given *n* positive integers, find the **largest 2** numbers in it.

Input Format

The *first* line of input contains integer '*n*'.

Then *next* n lines will contain an integer.

Constraints

$$2 \leq n \leq 100$$

Each integer is in the range $[1, 10000]$

Output Format

Print space-separated 2 integers denoting the 2 largest integers among the given integers. Print the 2nd largest integer and then the 1st largest integer.

Sample Input 0

```
5
1
5
2
4
3
```

Sample Output 0

```
4 5
```

Explanation 0

First line of input is 5. Next 5 lines contain an integer. Out of these, 4 and 5 are second largest and largest respectively. Hence, they've been printed out as required.

Sample Input 1

```
3
1
2
2
```

Sample Output 1

```
2 2
```

Explanation 1

3 in the first line denotes the number of inputs. out of the 3 values in the succeeding lines, 2 and 2 are respectively, the largest and the second largest numbers.